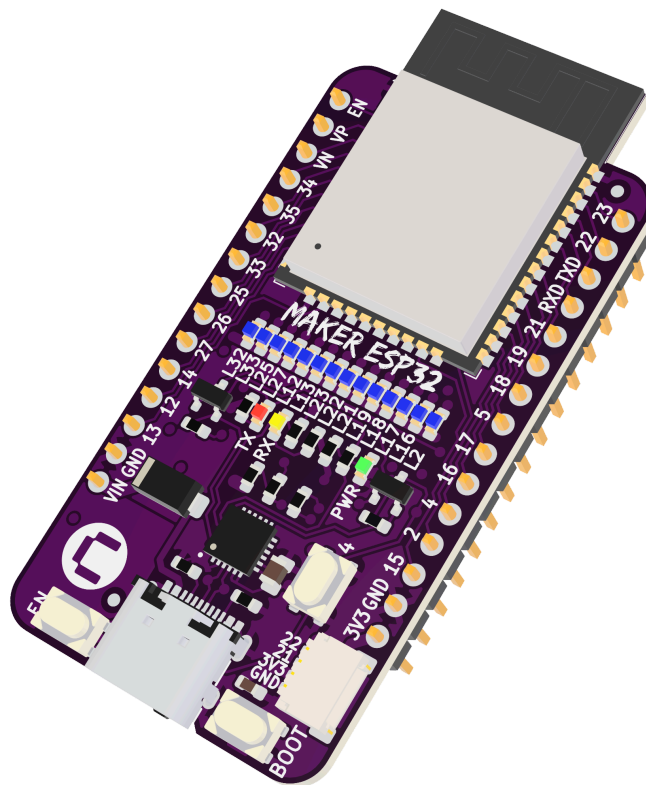




MAKER ESP32

Simplifying Digital Making



Datasheet

Rev 1.0
June 2026

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1. BOARD LAYOUT & FUNCTION

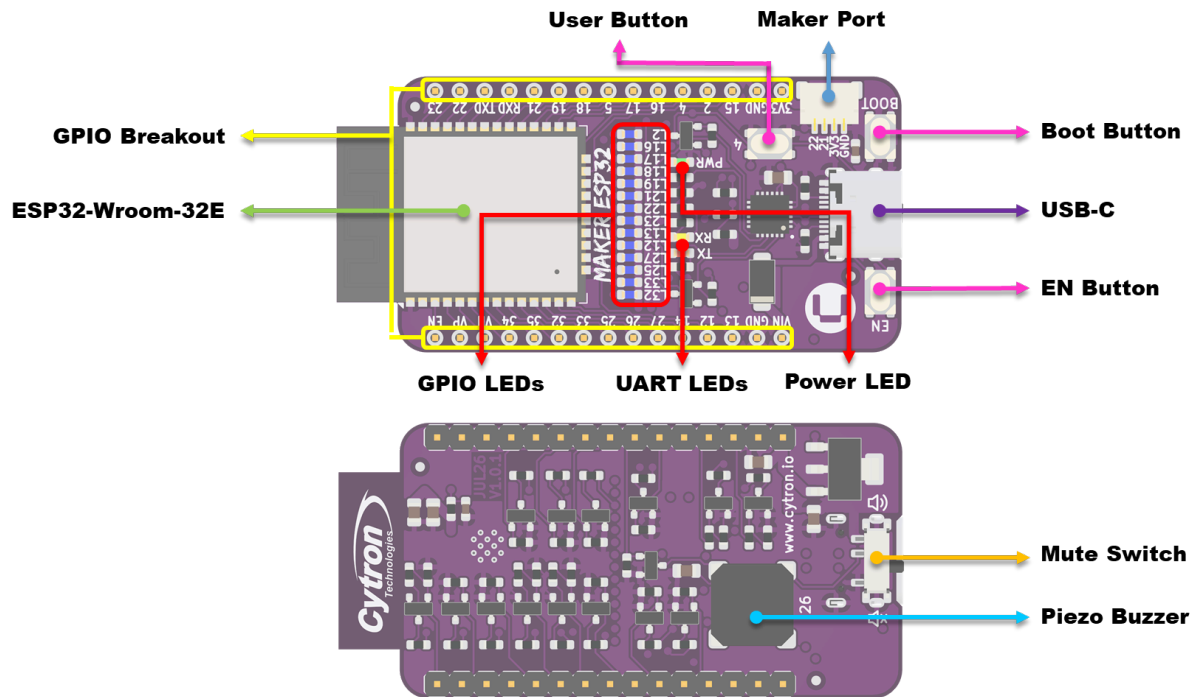
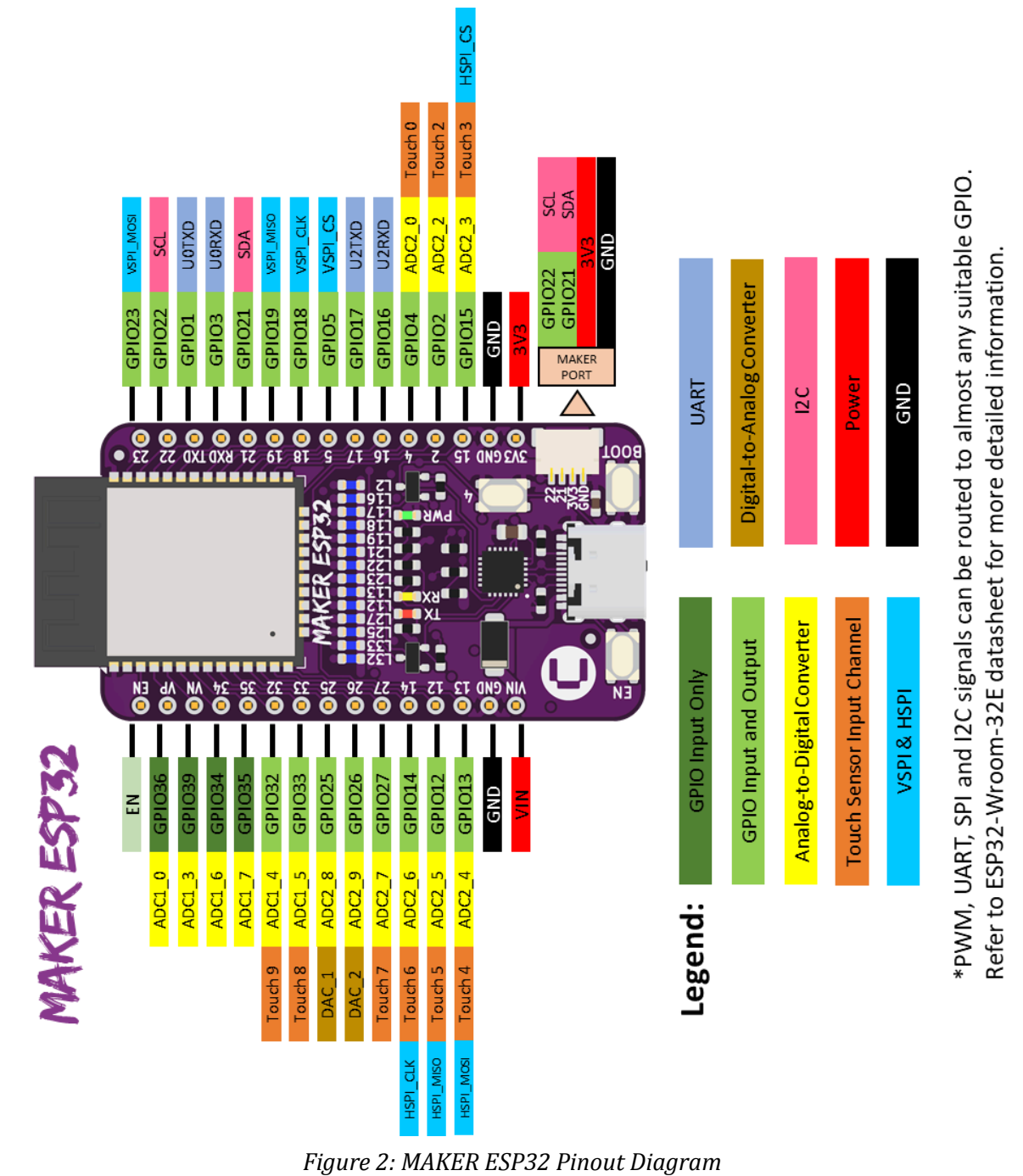


Figure 1: MAKER ESP32 Board Functions

Feature	Function
GPIO LEDs	LED indicators for ESP32 GPIOs. Turns on when the IO state is high. Connected to GPIO2, 12, 13, 16, 17, 18, 19, 21, 22, 23, 25, 27, 32 and 33.
Power LED (3.3V)	Indicates that the board is receiving power and operating normally.
UART LEDs	Indicate serial communication activity between the ESP32 and the host computer during data transmission and reception.
Boot Button	Places the ESP32 into firmware download mode when pressed during reset, allowing new firmware to be uploaded.
User Button	General-purpose push button connected to a GPIO 4 pin for custom user interactions and application control.
EN Button	Resets the ESP32 microcontroller and restarts the running application.
USB-C	Provides 5V power input and USB-to-UART communication for firmware upload and serial monitoring.
GPIO Breakout	GPIOs breakout for external connection. Compatible with NodeMCU ESP32 form factor.
Maker Port	JST-SH 4-Ways Connector for external modules. Compatible with Qwiic, STEMMA QT and Grove (Via Conversion Cable).
Mute Switch	Used to mute the buzzer.
Piezo Buzzer	Programmable passive piezo buzzer. Connected to GPIO26.
ESP32-WROOM-32E	ESP32-WROOM-32E-N8 module from Espressif. 8MB FLASH

Table 1: MAKER ESP32 Board Functions

2. PINOUT DIAGRAM



3. SPECIFICATIONS

No.	Parameters		Min	Max	Unit
1	Power Input Voltage		4.6	5.5	VDC
2	3V3 Maximum Current (Including Onboard Usage)		1.35		A
3	Digital Input Voltage	Low Level (V_{IL})	-0.3	0.8	V
		High Level (V_{IH})	2.5	3.6	V
4	Analog Input Voltage		0	3.3	V
5	Digital Output Voltage	Low Level (V_{OL})	0	0.4	V
		High Level (V_{OH})	2.6	3.3	V
6	Digital Output Current	Low Level Sink (I_{OL})	28		mA
		High Level Source (I_{OH})	40		mA
7	Internal Pull Resistor	Pull-Up (R_{PU})	45		k Ω
		Pull-Down (R_{PD})	45		k Ω
8	Operating Temperature		-40	85	°C

Table 2: MAKER-ESP32 Absolute Maximum Ratings

4. DIMENSION

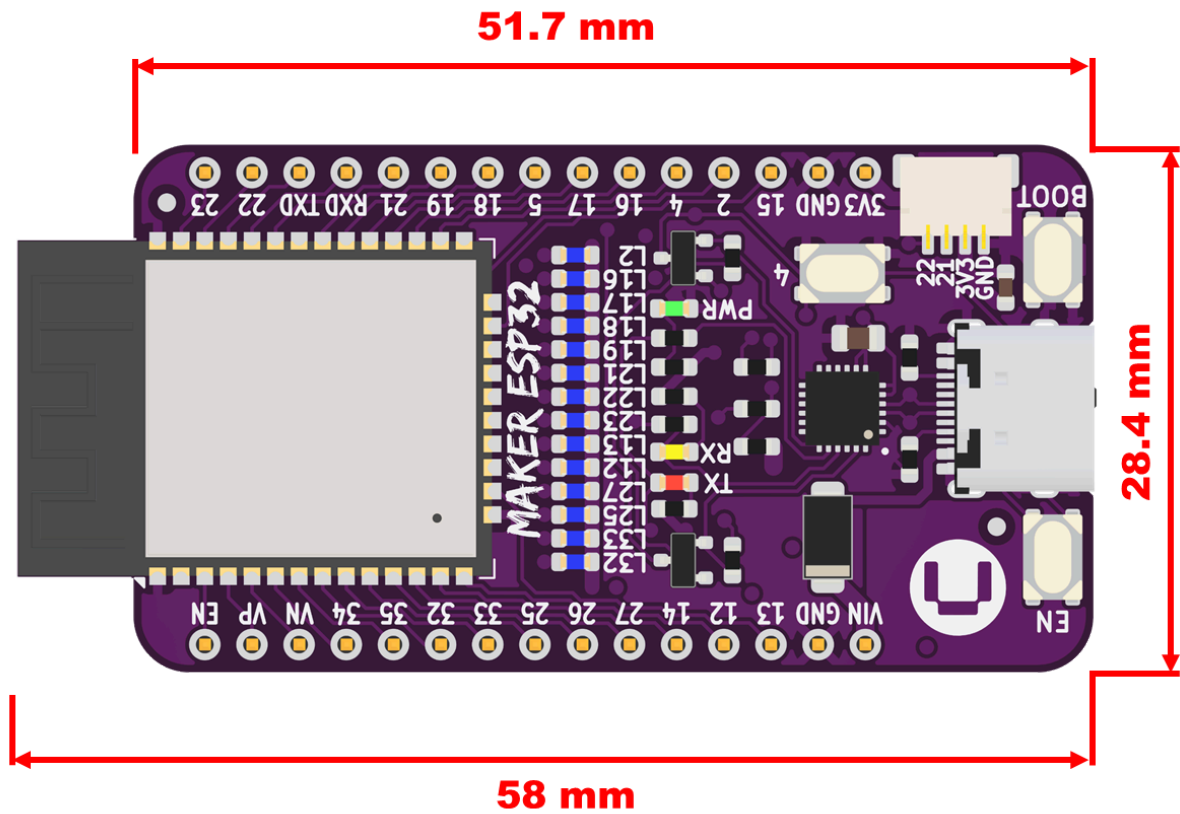


Figure 3: MAKER ESP32 Board Dimension

5. POWER TREE

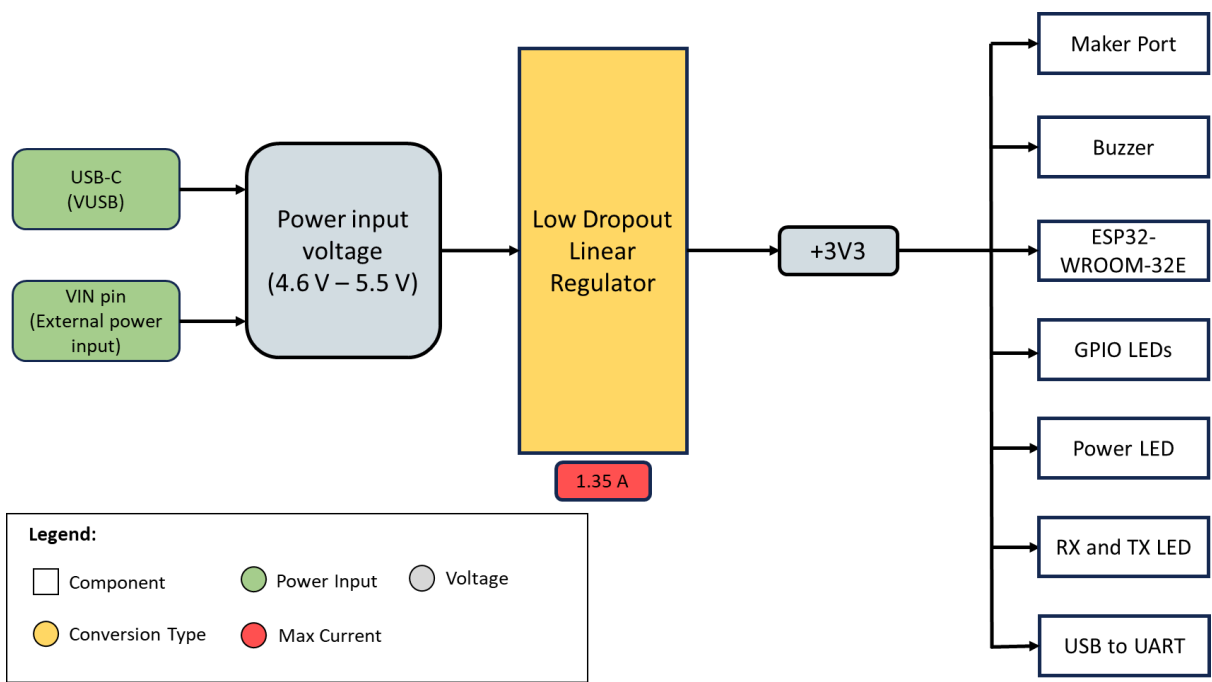


Figure 4: MAKER ESP32 Power Tree

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